

AI-Based Smart Grid Power Theft Detection System

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Abstract

Electricity theft and abnormal energy consumption remain major challenges for modern power distribution systems.

With the widespread deployment of smart meters in smart grid infrastructures, utilities now collect large volumes of electricity consumption data. While this data provides valuable insights for grid management, it also introduces the challenge of analyzing consumption patterns efficiently.

This report presents a machine learning-based framework for detecting abnormal electricity consumption behavior using simulated smart meter data. The system generates synthetic electricity usage data for multiple consumers and extracts statistical features such as average load, maximum load, and consumption variation. Machine learning techniques including Support Vector Machines (SVM), Artificial Neural Networks (ANN), and K-Means clustering are applied to analyze electricity usage patterns and identify suspicious consumption behavior.

In addition to machine learning analysis, an energy analytics dashboard is developed to visualize key consumption statistics across multiple consumers. The results demonstrate how data-driven methods can support utilities in monitoring electricity usage and identifying potential non-technical losses in smart grid environments.

Keywords

Smart Grid, Electricity Theft Detection, Machine Learning, Smart Meter Analytics, MATLAB, Energy Data Analysis

1. Introduction

The traditional electrical grid is evolving toward a smart grid architecture that integrates communication, automation, and advanced data analytics technologies. Smart meters play a crucial role in this transition by providing real-time electricity consumption data from consumers. This detailed monitoring enables utilities to analyze energy usage patterns, improve demand forecasting, and detect abnormal consumption behavior.

Electricity theft, however, continues to be a major concern for power utilities. Non-technical losses caused by electricity theft result in significant financial losses and may also lead to grid instability. Conventional detection methods rely on manual inspections and rule-based approaches that are often inefficient and unable to detect complex patterns in consumption data.

Machine learning techniques offer an effective solution for analyzing smart meter datasets and identifying abnormal patterns automatically. By learning the characteristics of normal electricity usage, machine learning models can detect deviations that may indicate electricity theft or other irregular behavior.

This project develops a MATLAB-based smart grid analytics system that simulates electricity consumption data and applies machine learning models to analyze consumer behavior. The system also provides visualization tools that allow utilities to interpret consumption patterns effectively.

The objectives of this project include:

- Simulating smart meter electricity consumption data for multiple consumers
- Extracting statistical features from electricity load profiles
- Applying machine learning models to detect abnormal consumption patterns
- Performing consumer behavior clustering using unsupervised learning techniques
- Developing visualization dashboards to analyze electricity usage patterns

2. Smart Meter Dataset Simulation

To evaluate the proposed system, a synthetic dataset representing smart meter electricity consumption was generated.

The dataset simulates electricity usage for multiple consumers over a time period representing daily energy usage.

Each consumer is represented by a time-series signal describing electricity consumption at different time intervals.

The generated load curves simulate realistic variations in electricity demand during different periods of the day.

Random noise is introduced to mimic real-world measurement variations.

The simulated dataset includes both normal and abnormal consumption patterns. Normal patterns represent typical electricity usage behavior, while abnormal patterns simulate possible electricity theft scenarios where the recorded energy usage may be artificially reduced or manipulated.

This dataset provides a controlled environment for evaluating the performance of machine learning models in detecting anomalous electricity consumption behavior.

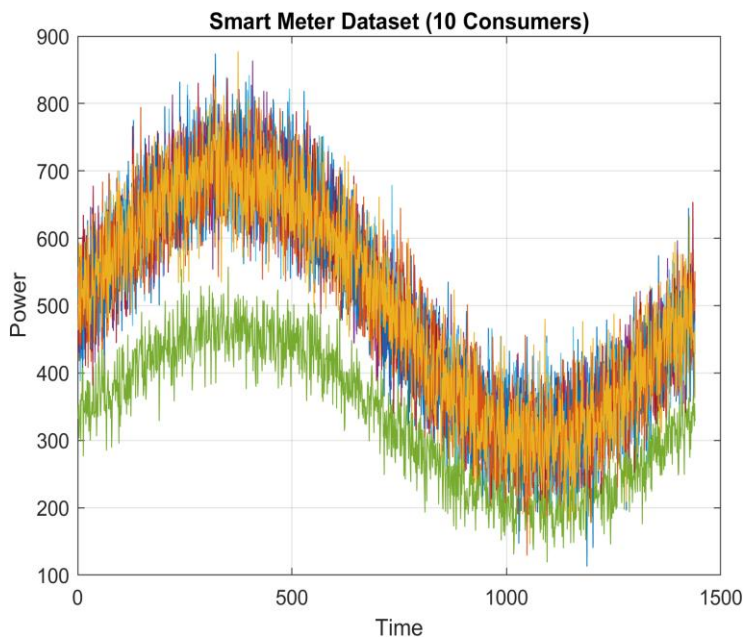


Figure 1: Simulated smart meter dataset showing electricity consumption patterns for multiple consumers.

3. Electricity Consumption Pattern Analysis

Analyzing electricity consumption patterns is essential for identifying abnormal usage behavior. In this project, consumption patterns are examined by comparing normal usage profiles with potential electricity theft patterns.

Normal electricity consumption typically follows a predictable daily pattern with peak usage during morning and evening hours and lower consumption during late night periods. These patterns are influenced by common household activities such as lighting, appliance usage, and heating or cooling systems.

In contrast, electricity theft patterns may exhibit irregular characteristics such as sudden drops in recorded consumption, inconsistent load patterns, or unusually low energy usage compared to expected behavior. Detecting these anomalies requires analyzing statistical properties of consumption data.

Visualization techniques are used to compare normal and abnormal consumption patterns and identify key differences between them.

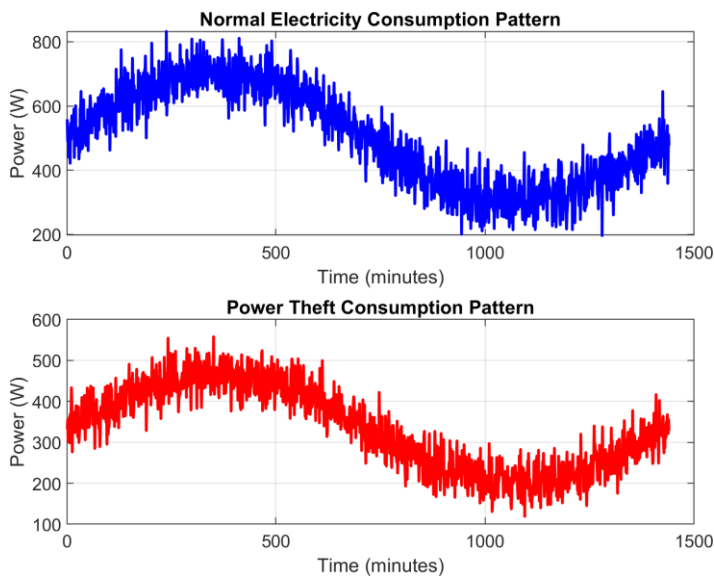


Figure 2: Comparison between normal electricity consumption and potential power theft patterns.

4. Consumer Clustering using Machine Learning

Unsupervised machine learning techniques can be used to analyze patterns in electricity consumption data without requiring labeled training samples. In this project, K-Means clustering is applied to group consumers based on electricity usage characteristics.

Two key features are extracted from each consumer's load profile:

- Average power consumption
- Maximum power consumption

These features provide a simple representation of electricity usage behavior. Consumers with similar consumption patterns are grouped into clusters using the K-Means algorithm.

Clustering allows utilities to identify groups of consumers with abnormal electricity usage behavior. For example, consumers with significantly lower average consumption compared to their maximum load may indicate potential electricity theft or irregular meter readings.

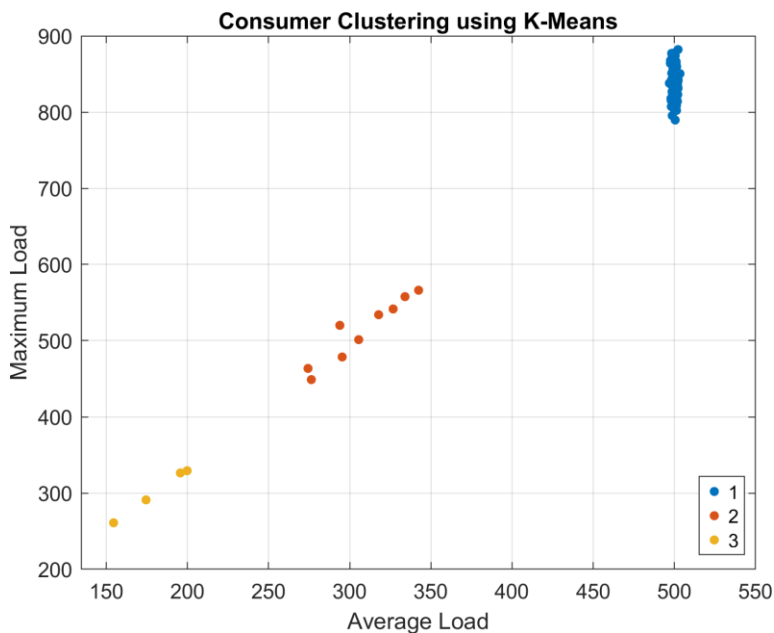


Figure 3: Consumer clustering results using K-Means algorithm.

5. Smart Grid Energy Analytics Dashboard

To support data interpretation, an energy analytics dashboard is developed to visualize electricity consumption statistics across multiple consumers. The dashboard provides several key insights including:

- Average electricity consumption per consumer
- Maximum load distribution
- Statistical distribution of electricity usage
- Consumption variation over time

These visualizations allow engineers and grid operators to quickly identify abnormal electricity usage patterns and evaluate overall energy consumption trends.

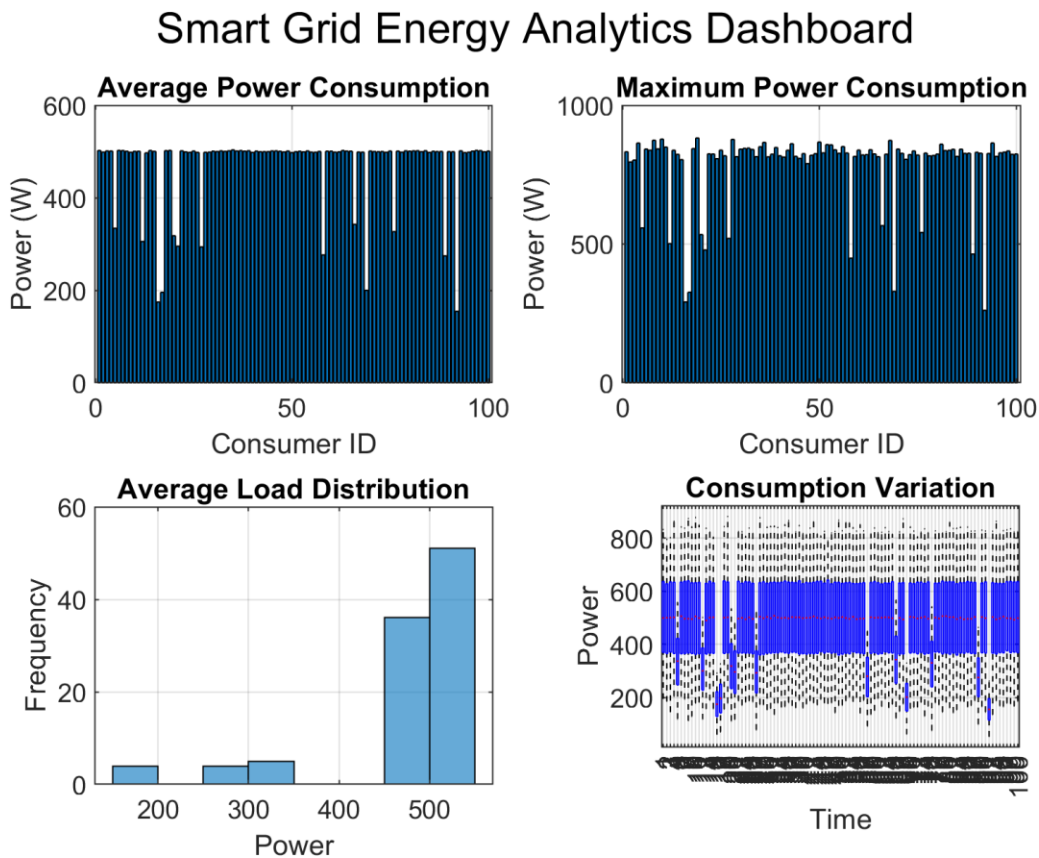


Figure 4: Smart grid energy analytics dashboard displaying electricity consumption statistics.

6. Discussion

The results demonstrate that machine learning techniques can effectively analyze electricity consumption patterns in smart grid environments. Feature extraction from load profiles provides valuable insights into consumer behavior, while clustering techniques allow the identification of abnormal usage patterns.

Although the dataset used in this project is simulated, the methodology can be applied to real-world smart meter datasets. With the increasing availability of smart grid data, machine learning models can significantly improve electricity theft detection and grid monitoring systems.

7. Conclusion

This project presents a machine learning–based approach for analyzing electricity consumption patterns and detecting abnormal behavior in smart grid systems. By combining smart meter data simulation, feature extraction, machine learning analysis, and visualization tools, the system demonstrates how intelligent data analytics can enhance monitoring and decision-making in modern power distribution networks.

The proposed framework highlights the potential of machine learning for improving electricity theft detection and energy monitoring in smart grids.

References

- [1] G. Chicco, "Overview and performance assessment of clustering methods for electrical load pattern grouping,"
Energy, vol. 42, no. 1, pp. 68–80.
- [2] A. Nizar, Z. Dong, and Y. Wang, "Power utility nontechnical loss analysis with extreme learning machine method,"
IEEE Transactions on Power Systems.
- [3] J. Gao et al., "Smart meter data analytics for electricity theft detection," IEEE Smart Grid Conference.
- [4] S. Amin and B. Wollenberg, "Toward a smart grid: Power delivery for the 21st century,"
IEEE Power and Energy Magazine.